



Michael Boiman

Freelance Quality Engineer & AI Architect • 20+ Years QA • 4+ Years AI Development

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● **Freelance** · available for new projects

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DB · DVAG · TÜV Süd

Enterprise clients

QA & legacy migration

EN 16931

E-invoicing in production

Peppol pipeline live (SAP ByDesign)

Live demo

AI agent

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Quality Engineering | LLM/AI Architecture | CI/CD Automation | Multi-Cloud | Peppol/E-Invoice

Career Profile

Quality Engineer and AI Automation Expert with 20+ years in software quality assurance and 4 years of specialized AI development. Quality engineering is not a closed chapter but an ongoing present — both tracks feed into each other.

In enterprise projects (**DB Vertrieb**, **DVAG**, **TÜV Süd**), quality monitoring systems took shape, AI-driven test automation was added, and legacy migrations were secured with 24/7 validation.

Today I combine quality engineering with AI architecture: MCP and A2A protocols, multi-agent systems, and e-invoicing to EU standards — bundled into one orchestration hub across 20+ repos, agents, and protocols.

I also pass this knowledge on — in developer and business workshops and as an invited guest speaker at TU Darmstadt — translating hands-on AI for technical and non-technical audiences alike.

PROJECTS & SOLUTIONS

Quality Dashboard - Real-time Overview

Grafana Elasticsearch Python CI/CD Playwright

Real-time quality overview for enterprise go/no-go decisions.

- **Problem:** Release decisions required hours of gathering data from 10 test tools and 5 environments.
- **Solution:** End-to-end pipeline with live dashboard, multi-source integration and automated PDF reports.
- **Result:** Data-driven release decisions in real time instead of manual collection.

24/7 Automated Legacy Migration Validator: Dual-Run Quality Engineering

Java Cucumber Kibana Graylog Elasticsearch

Dual-run validation for safely migrating a mainframe price calculation.

- **Problem:** Millions of price combinations had to migrate without quality loss or data errors.
- **Solution:** 24/7 validation with randomly generated tests, live dashboard and full deviation traceability.
- **Result:** Safe migration without data loss — all combinations validated around the clock.

Technical Skills

Quality Engineering Playwright

Cucumber/Gauge Python

Claude Code (Skills, Hooks, MCP)

LLMs (Claude, Gemini, GPT)

Agent Orchestration (MCP & A2A)

Google ADK LangChain

Peppol/E-Invoicing (EN 16931)

CI/CD Kubernetes

Azure Functions Grafana

Elasticsearch REST APIs

Docker JMeter Gatling

Agile (Scrum/Kanban)

Education

Diploma in Computer Science (UAS)

Cologne University of Applied Sciences, Gummersbach

2000 – 2005

ISTQB Certified Tester

2005

Certified Dynatrace Diagnostics Basic Training

2007

Languages

German Native

English Professional

E-Invoicing Automation Platform

Python Azure Functions Peppol Storecove SAP ByDesign

Production e-invoicing pipeline compliant with EU standard EN 16931.

- **Problem:** Inbound and outbound invoices in multiple formats (ZUGFeRD, XRechnung, UBL) required EN 16931-compliant processing.
- **Solution:** Peppol connection via Storecove, hexagonal architecture, automatic transfer to SAP ByDesign.
- **Result:** Production system — invoice processing runs automated end-to-end into the ERP.

AI Context Orchestrator — Multi-Repo Development Platform

Python YAML Claude Code Plugins Hooks API GitHub Actions

Central control system for AI-assisted development across 20+ repositories.

- **Problem:** Development across 20+ repos and multiple customers required constant context switching and manual coordination.
- **Solution:** Orchestration hub with ecosystem.yaml as single source of truth, 40+ skills and hook-based permission system.
- **Result:** Workflows from incident response to meeting documentation largely automated.

PROFESSIONAL EXPERIENCE

LLM Infrastructure Architect & Automation Engineer

06/2025 – present

BKS - AI Research & Development

Development of LLM-orchestrated Enterprise Development Ecosystem: Fully integrated system for knowledge management, project automation, and development workflows with Claude, Gemini, and OpenAI GPT.

Key Responsibilities

- **Open Source MCP Server Portfolio:** Development and publication of 3+ MCP servers on GitHub (bks-wiki-mcp, hubspot-mcp-bks, bks-codex) with uvx distribution, Claude Desktop integration, and multi-format support (Markdown, JSON, YAML)
- **Multi-Agent Orchestration System:** Implementation of production-ready A2A system with Orchestrator Agent (routing hub) and Knowledge Agent for inter-agent communication, capability-based Agent Cards, and JSON-RPC 2.0
- **Git Wiki Transformation:** Migration of Confluence knowledge base to structured Git-based wiki with hierarchical organization and multi-repository architecture (submodules for customer projects)
- **MCP Navigation Server:** Development of Model Context Protocol server (Python) with intelligent hierarchy navigation, automatic content discovery, and SharePoint integration

Tools & Technologies

- LLM Orchestration: Claude (primary), Google Gemini, OpenAI GPT, Model Context Protocol (MCP), Google Agent Development Kit (ADK)
- AI Development Platform: Claude Code, Plugin Architecture (Skills, Hooks, Commands, Agents), YAML/Markdown Configuration
- Multi-Agent: A2A Protocol, Orchestrator/Knowledge Agents, Agent Cards, JSON-RPC 2.0, SSE Streaming
- Backend: Python 3.13, FastAPI, uvx Distribution, Git Submodules
- Automation: GitHub Actions, GitHub GraphQL API, gh CLI
- Integration: SharePoint API, Clockify API, Elasticsearch, Natural Language Processing
- Development: Bash Scripting, jq, Docker, Multi-Agent Systems; orchestrated repos span Python, TypeScript, Rust and Go

Quality Engineering

09/2025 – present

TÜV Süd

Quality Engineering for an enterprise platform for AI-powered document processing in medical device certification (MDR/IVDR) with AI chat integration and data pipeline automation.

Key Responsibilities

- **BDD API Test Framework:** Development of comprehensive test framework with Python/Behave, self-healing authentication, HTML report generator with embedded API responses
- **E2E Test Automation:** Playwright E2E test suite for complete user journey (Setup -> Upload -> Transformation -> Download) with 500 error handling and route mocking
- **Performance & Monitoring:** Development of Azure Monitor Workbooks (QA Live Testing Companion, Error Investigation Assistant) with KQL for end-to-end monitoring
- **Bug Analysis & Quality Intelligence:** Implementation of comprehensive bug tracking system with WIQL queries, automated dashboards, and executive PDF reporting

Tools & Technologies

- Testing: Behave, Playwright, TypeScript, Python, pytest, Page Object Model, Fixtures
- Monitoring: Azure Monitor Workbooks, KQL, Application Insights, Log Analytics
- Backend: FastAPI, Python, Pydantic, SQLAlchemy, Azure Functions, Azure Service Bus
- Frontend: React, TypeScript, Vite, TanStack Router
- AI Integration: Azure OpenAI (GPT-4o), LangChain, Azure AI Search, Embeddings
- DevOps: Azure DevOps Pipelines, Git, Docker, Azure Blob Storage, Poetry

Technical Lead & AI Automation Architect

01/2024 – 04/2025

BKS

Technical lead & development: Design and implementation of a scalable automation solution for processing e-invoices and PDF invoices fetched from a mailbox of a leading online job platform (under NDA). Full responsibility from requirements analysis through implementation to production go-live.

Focus areas

- **Workflow automation:** Extraction, analysis and classification of incoming invoices using Python and Azure Functions.
- **E-invoice classification & data extraction:** Full processing in accordance with EU standard EN 16931 (ZUGFeRD, XRechnung, Factur-X), including structural and content parsing.
- **Peppol Network Integration:** Connection to European e-invoicing network via Storecove Access Point for automated invoice receipt and dispatch with UBL/CII formats.
- **SAP ByDesign Integration:** Automatic transfer of validated invoice data to SAP ByDesign for seamless ERP integration.

Tools & technologies

- Azure Functions: Serverless orchestration of automation workflows.
- Python: Core logic for data extraction, classification and integration with automated CI/CD processes, packaging, and comprehensive unit and integration tests for quality assurance.
- E-Invoicing: Peppol Network, Storecove API, ZUGFeRD/Factur-X, XRechnung, UBL 2.1, CII
- ERP Integration: SAP ByDesign, Salesforce
- Microsoft Graph API: Access to Outlook mailboxes and SharePoint libraries.
- GitHub: Version control, CI/CD pipelines and collaborative development.
- Kibana: Monitoring, log analysis and performance dashboards.
- Streamlit: For developing an interactive web application for visualization.
- AI Automation: Claude AI, Custom Skills Development, Git Analysis, Natural Language Processing
- Protocols: Model Context Protocol (MCP), Agent-to-Agent (A2A), JSON-RPC 2.0

Platform Quality Architect (CI/CD, Monitoring, Automation)

08/2021 – 05/2025

DVAG

Test management & coordination, as well as introduction of CI/CD with quality-gate setup. Acting as cross-team Quality Lead: onboarding, guidance, consulting and training for the introduction of quality metrics and test automation within the pipeline across teams.

Result: Real-time sprint-quality transparency via Grafana dashboards, quality-gated releases in the teams' CI/CD pipelines — data-driven release decisions replacing manual data gathering.

Tools

- IntelliJ / Visual Studio Code
- Java / JavaScript / TypeScript
- Playwright / Gauge / Karate
- Python for processing and analysis of test and reporting data
- CI/CD / GitHub – reusable workflows
- Helm / Containers
- JIRA / Confluence
- Grafana
- Behaviour-Driven Development
- Consumer-Driven Contracts (Pact.io)

AI-driven Automated QA Environment for Energy Infrastructure

07/2024 – 01/2025

AkkuSwap Startup

QA Leadership for EU-wide Battery Swap Infrastructure: AI-driven simulation tool for battery swap station network with energy infrastructure integration

Technical Stack

- Infrastructure: Linux, Docker, Azure OpenAI
- Automation: pytest, AI code generation integration patterns
- Monitoring: Grafana, Azure Monitoring
- Energy Systems: Battery swap infrastructure, grid integration simulation

Key Achievements

- Established QA methodology for AI-driven infrastructure simulation
- Validated proof-of-concept with measurable reliability improvements
- Created testing framework for energy sector critical systems

AI Engineering Lead & ML Solutions Architect

04/2023 – 04/2024

BKS on behalf of Ryze

Project management & technical lead: Development and deployment of an AI-driven email bot to automate customer communication, AI-assisted processing of incoming emails and triggering of follow-up activities in existing SAP systems. Responsible for end-to-end project coordination and team management from idea to implementation.

Tools & technologies

- Azure Functions: For serverless application architectures and ML model execution.
- Python: Main programming language for the email bot and ML models.
- Microsoft Graph API: For accessing and processing emails in Microsoft Outlook 365.
- SAP API: For querying customer data and communicating with SAP systems.
- GitHub: For version control, CI/CD pipelines and collaborative development.
- Kibana: For real-time monitoring and system performance analysis.
- Streamlit: For developing an interactive web application for system monitoring.

DevOps Quality Lead & Dashboard Architect

01/2017 – 05/2021

DB Vertrieb

Test management, test coordination, test automation – agile (Kanban, Scrum, SAFe): Coordination as PO of a cross-team QA team including cross-team onboarding, support, consulting, training

- Tools
- IntelliJ
 - Java
 - Cucumber / Cypress
 - JMeter / Gatling
 - CI/CD / GitLab CI / Jenkins
 - Helm / Docker
 - JIRA / Confluence
 - Kibana / Grafana / Instana / Graylog
 - Behaviour Driven Development
 - Consumer Driven Contracts (Spring Cloud Contract)
 - Ionic (hybrid app)
 - Selenium

Earlier Positions

09/2015 – 01/2017	Performance Engineering Lead & Test Architect — DB Systel	Load and performance tests and analysis – test manager, test designer, analyst
04/2015 – 09/2015	Senior iOS Developer & Mobile Architect — Telekom	Senior iOS Developer
05/2009 – 03/2015	Quality Assurance Lead & Test Automation Architect — Siemens	QA, test management, test automation, requirements engineering
03/2009 – 05/2009	Performance Test Engineer & Load Testing Specialist — ING-DIBA	Load and performance tester
01/2006 – 12/2008	Performance Test Engineer & Quality Consultant — British Telecom, Mobiliar, DB-Systel, Sparkassen Informatik, Loyalty Partner, Telekom, Itelium, Deutsche Post, Postbank	Load and performance tester / tester

WORKSHOPS & PRESENTATIONS

Guest Lecture: AI-Assisted Software Development in Practice
TU Darmstadt — Information Systems Group · Lab course “AI Startup: From Idea to Execution” — 04/2026

Workshop: Development with Generative Language Models
Developer Workshop - Agent-based Software Development — 06/2025

Presentation: Efficient Documentation through Automation
Enterprise Presentation - AI-powered Documentation Workflows — 04/2025

Applications of AI in Business Context
AI Workshop for Business Integration — 2023